SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0334 - MOBILE COMPUTING

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Name of the Student	**:MONISHA D**
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Duration	**:** 1 Week
Total Marks	**:** 50
Submission Date	**:** 03-11-2026

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).

- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.
- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.

- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

Answer:

Strategy 1: Fixed hexagonal frequency reuse (iterative cluster search)

The service region is represented by a hexagonal grid. A cluster contains N cells, and each cluster receives all 336 available duplex channels exactly once. Every cell receives approximately C/N channels. The same channel set is reused in other clusters at a distance D. The admissible hexagonal cluster sizes are $N = i^2 + ij + j^2$ for non-negative integers i and j, such as 1, 3, 4, 7, 9, 12, 13. N = 7 corresponds to (i,j) = (2,1).

Access and sharing: in an FDMA/TDMA system, a cell receives a fixed set of carrier/time-slot channels. Control channels carry broadcast, paging and access signalling; traffic channels carry calls. In an OFDMA system, a fixed subset of resource blocks can be assigned to each cell, with reuse coordinated between neighbouring cells.

Design principle: spatial reuse. It is investigated because it is simple, predictable and can be audited mathematically. Its weakness is that a large N reduces channels per cell, while a small N increases co-channel interference.

```
INPUT: total channels C=336; candidate indices (i,j); measured path-loss n
FOR each integer pair (i,j):
    N = i*i + i*j + j*j
    IF N is in the supported candidate set:
        channels_per_cell = floor(C/N)
        Q = sqrt(3*N)
        D = Q*R
```

```
        estimate first-tier SIR = (D/R)^n / 6
        retain N if SIR >= required_linear_SIR
    SELECT the smallest feasible N that meets SIR and blocking constraints
    assign N disjoint channel sets to the N cells in each cluster
    repeat the cluster pattern across the township
```

Strategy 2: Sectorised frequency reuse

Each macrocell is divided into directional sectors, typically three 120° sectors or six 60° sectors. Directional antennas concentrate transmit and receive energy into a sector and reduce the number of dominant co-channel interferers seen by a mobile. The frequency reuse cluster may remain N = 7, but the effective first-tier interferer count is reduced. The idealised approximation uses six interferers for an omnidirectional cell, two dominant interferers for 120° sectoring, and one dominant interferer for 60° sectoring. Actual patterns depend on antenna front-to-back ratio, azimuth, downtilt and side lobes.

Access and sharing: the channel pool is divided among sectors or assigned with coordinated reuse. Control channels must remain available for access and mobility in each sector. Sectorisation does not create additional spectrum; it improves spatial filtering and may allow a smaller reuse distance or more robust SIR.

Design principle: spatial filtering. It deserves investigation because academic blocks and the arterial road produce directional traffic corridors, and antenna patterns can suppress interference without adding spectrum.

```
    FOR each macrocell:
        divide coverage into S=3 (120-degree) or S=6 (60-degree) sectors
        measure/estimate sector azimuth and antenna pattern
        assign channel groups to sectors using neighbour conflict constraints
        compute SIR using only geometrically visible dominant interferers
        if sector SIR and blocking meet targets:
            retain sector plan
        else:
            increase reuse distance or add a microcell
```

Strategy 3: Hierarchical macro–micro layer with dynamic channel assignment (recursive)

A macro layer provides continuous coverage and mobility; micro/pico cells are added in high-traffic locations such as lecture halls, hostels, libraries and the stadium. A cell is split recursively when its measured busy-hour traffic exceeds its engineered capacity. Dynamic channel assignment (DCA) or channel borrowing allows a lightly loaded cell to lend an idle channel to a congested neighbour, provided interference and reuse constraints are satisfied. This is distinct from simply changing N: it changes the network hierarchy and assignment process.

Access and sharing: macro cells provide common control and mobility coverage; microcells provide local traffic capacity. A controller selects a channel/resource block from the available pool using load, interference and neighbour occupancy. In LTE/5G-style OFDMA, this is coordinated scheduling/resource-block allocation rather than a permanent voice-channel assignment.

Design principle: hierarchical spatial reuse plus trunking. It is investigated because traffic is highly non-uniform and grows over time. Its disadvantages are higher planning, backhaul, synchronisation and handoff complexity.

```
PROCEDURE PLAN(cell, demand):
    IF demand(cell) <= engineered_capacity(cell):
        keep cell
        RETURN
    deploy four child cells with radius R/2
    distribute the parent traffic among the children
    for each child:
        assign channels by DCA subject to neighbour interference constraints
        PLAN(child, child_demand)
END PROCEDURE
```

AT runtime:

when a cell requests a channel:

search its own free channels

then search permitted neighbour channels

borrow only if co-channel conflict and blocking constraints remain safe

otherwise queue/block the request

Preprocessing and cost responsibility

All strategies require a GIS traffic map (student population, building occupancy, road flows and stadium events), drive/walk tests, propagation and interference measurements, terrain/building data, candidate-site surveys, backhaul/power checks, and a hex-grid/site layout. The mobile operator bears the engineering, survey, site-acquisition and radio-planning costs. The university or township authority may provide building plans, access permissions and event schedules, but these are not free engineering inputs. The cost model must include survey labour, planning software, site rental, civil works, radios, antennas, power, backhaul, optimisation and periodic re-surveys.

TASK 2 – Identify the Governing Design Parameter **2 Marks**

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

Answer:

Strategy	**Dominant parameter**	**Reason and growth response**
Fixed reuse	Cluster size N (equivalently $Q=D/R=\sqrt{(3N)}$); channels/cell C/N	N controls the trade-off between co-channel interference and per-cell channel count. As demand rises, retain the smallest N that satisfies SIR; add cells rather than arbitrarily reducing N below the interference limit.
Sectorised reuse	Effective interferer count I and sector beamwidth; also N	Sectoring changes the number/strength of visible interferers. More directional sectors improve SIR; N still controls the channel pool per reuse cell. Growth may justify 60° sectors in hotspots.
Macro–micro + DCA	Cell radius R and offered load per cell A_cell; also free-channel availability	R controls the number of cells and spatial reuse; DCA responds to instantaneous load. As demand rises, R decreases in hotspots and more microcells are added selectively.

The governing parameter must be linked to both capacity and interference. N alone cannot describe a hierarchical design because two cells with the same N may have very different loads and radii. Therefore the adaptive design is governed primarily by the per-cell offered load relative to the Erlang-B capacity, while R determines how many such cells are needed.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km².
- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

Answer:

3.1 Hexagonal geometry and reuse distance

For a regular hexagon with centre-to-vertex radius R, the hexagon consists of six equilateral triangles of side R. Its area is $A_{cell} = 6(\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 = 2.598R^2$ km². A cluster of N cells has area $N A_{cell}$. Ignoring boundary inefficiency, the number of cells required for area A_T is $K = \text{ceil}[A_T/(2.598R^2)]$. In practice, add a

boundary/site-planning margin because the township is not a perfect hexagon and coverage overlaps.

For a hexagonal frequency plan, moving i cells along one axis and then j cells after a 60° turn reaches the nearest co-channel cell. The squared centre separation is $D^2 = 3R^2(i^2+ij+j^2)$. Since $N=i^2+ij+j^2$, $D=R\sqrt{(3N)}$, and reuse ratio $Q=D/R=\sqrt{(3N)}$.

N	Valid (i,j) example	Q=√(3N)	Channels/cell floor(336/N)
3	(1,1)	3.000	112
4	(2,0)	3.464	84
7	(2,1)	4.583	48
12	(2,2)	6.000	28

3.2 SIR derivation

Under a path-loss model, received power from a base station at distance d is proportional to d^(−n). For a mobile near the cell centre, the desired base station is at approximately R (a conservative cell-edge reference), and six first-tier co-channel base stations are each approximated at distance D. Thus SIR ≈ R^(−n)/[6D^(−n)] = (D/R)^n/6 = (3N)^(n/2)/6. In decibels, SIR_dB = 10 log10[(3N)^(n/2)/I], where I=6 for the omnidirectional approximation.

For sectoring, replace I=6 by an idealised I=2 for 120° sectors or I=1 for 60° sectors, while recognising that real antenna patterns produce non-zero side-lobe interference. This simplified expression is a planning approximation, not a substitute for site-specific propagation simulation.

N	SIR n=2.5, I=6	SIR n=3.5, I=6	SIR n=4, I=6
3	7.7 dB	11.2 dB	13.0 dB
4	8.8 dB	12.8 dB	14.8 dB
7	12.4 dB	16.6 dB	18.7 dB
12	15.6 dB	20.4 dB	22.9 dB

The supplied sanity check is reproduced: for N=7 and n=4, SIR=(21)^2/6=73.5, hence 10log10(73.5)=18.66 dB ≈18.7 dB. For n=2.5 the same reuse plan gives about 12.4 dB, meeting the selected 12 dB digital target but not an 18 dB target. In real networks the nearest interferers, shadowing and antenna patterns must be included.

3.3 Erlang-B traffic capacity

For m traffic channels and offered traffic A Erlangs, the Erlang-B blocking probability is B(0,A)=1 and B(m,A)= [A B(m−1,A)]/[m + A B(m−1,A)]. This recurrence is numerically stable. For each m, increase A until B(m,A)=0.02. With N=7, m=floor(336/7)=48 channels/cell; the 2% blocking traffic capacity is approximately 38.39 Erlangs per cell. The capacity values below are computed using the recurrence, not a linear channels-times-subscriber estimate.

N	m channels/cell	Erlangs/cell at 2% GoS (approx.)
3	112	99.62
4	84	72.53
7	48	38.39
12	28	20.15

For N=7, total carried offered traffic capacity across K cells is approximately C_sys(K)=K×38.39 Erlangs, assuming each cell has its own 48-channel group and enough reuse clusters are deployed. The total spectrum is reused spatially; this does not mean 48×K unique licensed channels. At the subscriber traffic rate 0.025 Erlang, one N=7 cell supports about 38.39/0.025 ≈ 1,536 subscribers at 2% blocking under the Erlang-B assumptions.

3.4 Worked township cell counts and capacity

With R_0=0.45 km, A_cell=2.598×0.45²=0.526 km² and K_0=ceil(12/0.526)=23 macrocells. With R_1=0.225 km, A_cell=0.1317 km² and K_1=ceil(12/0.1317)=92 microcells. These are idealised area counts; actual deployment needs boundary overlap and site constraints. At N=7, capacity is 23×38.39=882.97 Erlangs before splitting and 92×38.39=3,531.9 Erlangs after one uniform split.

Subscribers	Busy-hour offered traffic	23-cell capacity	92-cell capacity	Implication
1,000	25 E	883 E	3,532 E	Macro layer has ample capacity

10,000	250 E	883 E	3,532 E	Macro layer is sufficient in aggregate; local hotspots may need microcells
20,000	500 E	883 E	3,532 E	Macro layer remains sufficient in aggregate; check local distribution
100,000	2,500 E	883 E	3,532 E	One uniform split gives sufficient aggregate capacity at 2% cell GoS

For 100,000 subscribers, 2,500 Erlangs / 23 = 108.7 Erlangs per macrocell, far above the 38.39 Erlang cell limit. After one four-way split, 2,500/92 = 27.17 Erlangs per child cell, below the 38.39 Erlang 2%-blocking limit. Thus the 92-cell layout is a feasible first-order capacity plan.

3.5 Cost–capacity relation

Let S be the one-time survey and planning cost, c_s the annualised cost per deployed site (radio, antenna, rent, power and backhaul), and c_o the operating cost per carried Erlang. A simple annualised total cost is C_total(K)=S + K c_s + c_o A_carried. For a fixed-reuse network, K is driven by area and R, while the traffic capacity is K T_B(C/N), where T_B(m) is the Erlang-B offered load at 2% blocking. Hence the cost per supported Erlang is [S+Kc_s]/[K T_B(⌊C/N⌋)]. Sectoring adds antenna/installation/optimisation cost c_sec per sector but can reduce the required N or improve outage performance. Hierarchical DCA adds controller/backhaul/coordination cost c_DCA and selective microcell site cost, but avoids deploying small cells in low-demand areas. These terms must be populated with local quotations; inventing rupee amounts without site/vendor data would be misleading.

Fixed-reuse capacity per cell falls as N increases because m≈C/N. Under the idealised SIR model, SIR rises with N as N^(n/2). Thus the economic optimum is the smallest N that satisfies the SIR target and local blocking constraints, with site count adjusted to demand.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

Answer:

4.1 SIR by mobile position

The six-equidistant-interferer approximation is a useful reference but is not exact for every mobile position. Near the base station, the desired path is short, so desired received power rises and SIR is generally best. At mid-cell, the desired distance and interferer distances are intermediate. At the cell edge or corner, the desired link is weakest and one or more co-channel interferers may be relatively close; this is normally

the worst SIR location. Hexagonal corners are especially important because the mobile can be near two neighbouring cell boundaries.

A simple numerical illustration for N=7, n=4 uses the centre-to-first-tier distance D=4.583R. At a cell-centre mobile with desired distance 0.2R, the desired-to-interference ratio under six equal-distance interferers is (D/0.2R)^4/6 = about 3.27×10^5, or 55.1 dB. At the conservative edge reference desired distance R it is 18.7 dB. If the mobile is at a corner, the actual six interferer distances must be calculated individually; a close interferer can reduce SIR below the equal-distance estimate. Therefore 18.7 dB is a planning benchmark, not a guarantee at every point.

Shadowing adds a random attenuation term in dB. A 6–8 dB shadow fade on the desired link can reduce SIR by the same amount; a shadowed interferer may improve it. Increasing n from 2.5 to 4 increases the relative attenuation of distant interferers and raises the ideal SIR for fixed N, but dense buildings can also weaken the desired link and create coverage holes. Therefore a larger n is not automatically beneficial.

4.2 Blocking by traffic period

Blocking is lowest off-peak, rises at average load, and is highest in the busy hour or event surge. Erlang-B assumes Poisson call arrivals, independent call holding times, a loss system with no waiting queue, and a statistically stable offered load. For N=7, m=48, the design point is A≈38.39 Erlangs for 2% blocking. At A=27.17 Erlangs per cell, blocking is below 2%; at A=38.39 it is 2%; and above that it rises rapidly. A traffic burst lasting four hours can be handled by temporary capacity if it is predictable; the steady-state busy-hour assumption alone does not capture short spatially concentrated peaks.

Best-case joint condition: mobile near the serving base station and off-peak traffic. Average case: typical mid-cell position, average busy-hour traffic and median shadowing. Worst-case: cell corner, shadowed desired path, high interference and event-hour offered load. The SIR and blocking worst cases are related but not identical: interference can cause radio-link outage even when a traffic channel is free, while blocking can occur in a low-interference cell if all channels are occupied.

TASK 5 – Scaling and Sensitivity Analysis **3 Marks**

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

Answer:

For fixed spectrum C, channels per cell $m \approx C/N$; increasing N improves ideal SIR but reduces the number of channels available in each cell.

For the six-interferer model, $SIR_linear \propto (3N)^{(n/2)}$, so $SIR \propto N^{(n/2)}$; in dB, SIR_dB increases by $5n \log10(N_2/N_1)$ dB when N changes from N_1 to N_2.

For fixed coverage area, cell count $K \approx A_T/(2.598R^2)$, hence $K \propto 1/R^2$. Halving R creates approximately four times as many cells.

At a fixed cluster size, aggregate network capacity scales approximately with K, provided each cell can be assigned a full reuse set through spatial reuse and interference remains acceptable.

Subscriber demand grows from 25 Erlangs at 1,000 users to 2,500 Erlangs at 100,000 users. The 23-macrocell N=7 layout has an aggregate 883 Erlang 2%-blocking capacity

and is not sufficient at 100,000 users. A four-way split to 92 cells raises aggregate capacity to about 3,532 Erlangs. For demand above 100,000, further selective splitting or additional layers are needed.

A fixed N=7 omnidirectional plan is marginal for n=2.5 at 12.4 dB and fails an 18 dB requirement; it is suitable for the chosen 12 dB digital target. At n=4 it yields 18.7 dB under the ideal model. N=12 improves SIR but cuts channels/cell to 28 and Erlang capacity to about 20.15 E; it may therefore require more cells. Sectoring becomes attractive where interference, rather than channel occupancy, is the limiting factor. Hierarchical microcells become preferable once local demand exceeds a macrocell's 38.39 Erlang engineered capacity or when a small hotspot is surrounded by lightly loaded cells.

Break-even decision: retain macrocells while every cell's busy-hour demand remains below 38.39 E and radio coverage meets the target. Split selectively when a cell exceeds that load or its 5th-percentile SIR/outage violates the target. For the township-wide 100,000-subscriber forecast, one uniform split is the conservative first-order plan; in deployment, traffic mapping should favour selective splitting to reduce site cost.

Answer:

The base layout has $K_0=23$ macrocells of radius $R_0=0.45$ km. Each overloaded parent cell is divided into four approximately equal-area child cells of radius R/2. This follows area scaling: $(R/2)^2=R^2/4$, so four children approximately cover one parent. Let f_k be the fraction of cells split at round k, where $0\le f_k\le 1$.

Uniform splitting: K_k=4K_(k−1), R_k=R_0/2^k, and total idealised area coverage remains constant because $K_k\times 2.598R_k^2=K_0\times 2.598R_0^2$. If every cell is split, the total offered traffic capacity becomes $C_k=4^k C_0$, assuming each child receives the same 48-channel N=7 reuse allocation and sufficient interference coordination. For selective

splitting, K_k=K_(k−1)[1+3f_k], because each selected parent is replaced by four children, a net gain of three cells.

Selective capacity recurrence: C_k=C_(k−1)[1+3f_k] under equal per-cell capacity T_B=38.39 E and ideal channel reuse. If only a fraction f_k of cells is overloaded, demand must be redistributed to the new children; the recurrence is a network aggregate approximation, not a claim that every child carries identical traffic.

Transmit power: to maintain a comparable received power at half the cell radius under path loss exponent n, P′/P=(R′/R)^n=(1/2)^n, so P_k=P_0/2^(nk). The problem statement writes P′=P/2^n; this is the one-round result. In practice power is tuned to coverage, antenna height, penetration and interference, so this is a first-order scaling rule.

Handoff load: with average user speed v and characteristic cell-crossing length proportional to R, handoff rate per moving user is approximately proportional to v/R. Thus halving R approximately doubles per-user handoff frequency; after k uniform rounds it scales as 2^k. The base case is k=0: K_0=23, R_0=0.45 km, C_0=23×38.39=882.97 E.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round k − 1?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius R → R/2).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power (P′ = P/2^n), derived from your own splitting procedure.

- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

Answer:

The base layout has $K_0=23$ macrocells of radius $R_0=0.45$ km. Each overloaded parent cell is divided into four approximately equal-area child cells of radius R/2. This follows area scaling: $(R/2)^2=R^2/4$, so four children approximately cover one parent. Let f_k be the fraction of cells split at round k, where $0 \le f_k \le 1$.

Uniform splitting: K_k=4K_(k−1), $R_k=R_0/2^k$, and total idealised area coverage remains constant because $K_k \times 2.598 R_k^2 = K_0 \times 2.598 R_0^2$. If every cell is split, the total offered traffic capacity becomes $C_k=4^k C_0$, assuming each child receives the same 48-channel N=7 reuse allocation and sufficient interference coordination. For selective splitting, K_k=K_(k−1)[1+3f_k], because each selected parent is replaced by four children, a net gain of three cells.

Selective capacity recurrence: C_k=C_(k−1)[1+3f_k] under equal per-cell capacity T_B=38.39 E and ideal channel reuse. If only a fraction f_k of cells is overloaded, demand must be redistributed to the new children; the recurrence is a network aggregate approximation, not a claim that every child carries identical traffic.

Transmit power: to maintain a comparable received power at half the cell radius under path loss exponent n, $P'/P=(R'/R)^n=(1/2)^n$, so $P_k=P_0/2^{(nk)}$. The problem statement writes $P'=P/2^n$; this is the one-round result. In practice power is tuned to coverage, antenna height, penetration and interference, so this is a first-order scaling rule.

Handoff load: with average user speed v and characteristic cell-crossing length proportional to R, handoff rate per moving user is approximately proportional to v/R. Thus halving R approximately doubles per-user handoff frequency; after k uniform rounds it scales as 2^k. The base case is k=0: $K_0=23$, $R_0=0.45$ km, $C_0=23 \times 38.39=882.97$ E.

TASK 7 – Solve the Recurrence and Interpret the Result **4 Marks**

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

Answer:

For uniform splitting, K_k=4K_(k−1), K_0=23. Repeated substitution gives $K_1=4K_0$, $K_2=4(4K_0)=4^2K_0$, and therefore K_k=$K_0$4^k. Similarly C_k=$C_0$4^k. To support demand D, require $C_0$4^k≥D. Dividing by C_0 and taking logarithms gives k≥$\log_4(D/C_0)$, hence k_min=ceil[$\log_4(D/C_0)$].

For D=2,500 E and C_0=882.97 E, D/C_0=2.831 and $\log_4(2.831)$=0.751; therefore k_min=1. After one split, K_1=92, R_1=0.225 km and C_1=3,531.9 E. The design has a nominal aggregate margin of 1,031.9 E, although local uneven traffic can still require selective additions.

For selective splitting, the closed form is C_k=C_0∏_(r=1 to k)(1+3f_r), and K_k=K_0∏_(r=1 to k)(1+3f_r), under the equal-capacity approximation. The minimum number of selective rounds is the smallest integer k for which C_0∏(1+3f_r)≥D. If a constant fraction f is split each round, k_min=ceil{ln(D/C_0)/ln(1+3f)}. This reduces cost

when only a small fraction of cells is overloaded, but it requires accurate traffic maps and repeated optimisation.

The solution shows the cost–mobility trade-off: site count grows by four in a uniform split, while per-site transmit power falls by 2^n (one split) and handoff rate per moving user roughly doubles. Selective splitting avoids unnecessary sites and handoffs in quiet parts of the township.

TASK 8 – Implement and Test the Planning Strategies **4 Marks**

Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

Answer:

The following Python program is a reproducible reference simulator. It implements Erlang-B, valid cluster-size enumeration, hexagonal cell coordinates, first-tier Monte-Carlo SIR samples, and a simplified strategy comparison. It uses a fixed random

seed so the same mobile positions and shadowing samples are used across strategies. The first-tier SIR model is intentionally stated as an approximation: a full engineering simulator should import actual antenna patterns, building clutter, measured path-loss maps and all co-channel sites.

Implementation:

```
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import math
import numpy as np

SEED = 2026
CHANNELS = 336
TARGET_B = 0.02

def erlang_b(m, A):
    b = 1.0
    for i in range(1, m + 1):
        b = A * b / (i + A * b)
    return b

def erlang_capacity(m, target=TARGET_B):
    lo, hi = 0.0, max(1.0, 3.0*m)
    for _ in range(100):
        mid = (lo + hi) / 2
        if erlang_b(m, mid) > target:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2

def valid_clusters(limit=20):
    vals = {}
    for i in range(limit + 1):
        for j in range(limit + 1):
            n = i*i + i*j + j*j
            if 1 <= n <= limit:
                vals[n] = (i, j)
    return sorted(vals.items())

def hex_centres(rows, cols, R):
    # Pointy-top axial layout; nearest-neighbour spacing is sqrt(3)*R.
    pts = []
    for q in range(cols):
        for r in range(rows):
            x = math.sqrt(3)*R*(q + r/2)
            y = 1.5*R*r
            pts.append((x, y))
    return np.asarray(pts)

def first_tier_sir(N, n, samples=20000, sectors=1, seed=SEED):
    # Six idealised co-channel interferers at distance D=sqrt(3N)*R.
    # Random user positions are within a circular approximation of the serving cell.
    rng = np.random.default_rng(seed)
    R = 1.0
```

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```
    for a,b in zip(ix,iy):
        d = np.sqrt((x-a)**2+(y-b)**2)
        interf += d**(-n)
    # Ideal sector filtering: approximate removal of non-visible first-tier sites.
    # sectors=1 -> 6; sectors=3 -> 2; sectors=6 -> 1 dominant interferer.
    keep = {1:6, 3:2, 6:1}[sectors]
    # Retain the strongest 'keep' interferers per mobile.
    contributions = []
    for a,b in zip(ix,iy):
        d = np.sqrt((x-a)**2+(y-b)**2)
        contributions.append(d**(-n))
    strongest = np.sort(np.vstack(contributions), axis=0)[-keep:,:]
    interf = strongest.sum(axis=0)
    sir_db = 10*np.log10(desired/interf)
    return {
        "mean_dB": float(np.mean(sir_db)),
        "p5_dB": float(np.percentile(sir_db, 5)),
        "outage_pct_below_12dB": float(100*np.mean(sir_db < 12)),
        "outage_pct_below_18dB": float(100*np.mean(sir_db < 18))
    }

def compare(subscribers=(1000,10000,20000,100000)):
    rows = []
    for N in (3,4,7,12):
        m = CHANNELS // N
        cap = erlang_capacity(m)
        for users in subscribers:
            offered = users*0.025
            # Example 23 macro sites; after one uniform split 92 sites.
            for sites in (23,92):
                per_cell = offered/sites
                rows.append((N, users, sites, m, cap, per_cell,
                             erlang_b(m, per_cell)))
    return rows

if __name__ == "__main__":
    print("Valid clusters:", valid_clusters())
    for n in (2.5,3,4):
        for N in (3,4,7,12):
            for s in (1,3,6):
                print("SIR", N, n, "sectors", s,
                      first_tier_sir(N,n,sectors=s))
    for row in compare():
        print("N,users,sites,m,capacity,load/cell,blocking:", row)
    print("Sanity N=7,n=4:",
          10*math.log10((3*7)**(4/2)/6), "dB")
```

Output:




```
N,users,sites,m,capacity,load/cell,blocking: (3, 10000, 23, 112, 99.62382110612876, 10.869565217391305, 1.0957525660646094e-71)
N,users,sites,m,capacity,load/cell,blocking: (3, 10000, 92, 112, 99.62382110612876, 2.7173913043478826, 1.4107095734438536e-135)
N,users,sites,m,capacity,load/cell,blocking: (3, 20000, 23, 112, 99.62382110612876, 21.73913043478261, 1.0826293986863974e-42)
N,users,sites,m,capacity,load/cell,blocking: (3, 20000, 92, 112, 99.62382110612876, 5.434782608695652, 4.837813023386688e-103)
N,users,sites,m,capacity,load/cell,blocking: (3, 100000, 23, 112, 99.62382110612876, 108.69565217391305, 0.0553489984933768)
N,users,sites,m,capacity,load/cell,blocking: (3, 100000, 92, 112, 99.62382110612876, 27.17391304347826, 3.373690014707295e-34)
N,users,sites,m,capacity,load/cell,blocking: (4, 1000, 23, 84, 72.52882821734931, 1.0869565217391304, 1.1204160237714132e-124)
N,users,sites,m,capacity,load/cell,blocking: (4, 1000, 92, 84, 72.52882821734931, 0.2717391304347826, 6.766817353097692e-175)
N,users,sites,m,capacity,load/cell,blocking: (4, 10000, 23, 84, 72.52882821734931, 10.869565217391305, 6.321881284431123e-45)
N,users,sites,m,capacity,load/cell,blocking: (4, 10000, 92, 84, 72.52882821734931, 2.717391304347826, 5.864773046708789e-92)
N,users,sites,m,capacity,load/cell,blocking: (4, 20000, 23, 84, 72.52882821734931, 21.73913043478261, 2.326878877914632e-24)
N,users,sites,m,capacity,load/cell,blocking: (4, 20000, 92, 84, 72.52882821734931, 5.434782608695652, 7.492431786831192e-68)
N,users,sites,m,capacity,load/cell,blocking: (4, 100000, 23, 84, 72.52882821734931, 108.69565217391305, 0.2517907796227243)
N,users,sites,m,capacity,load/cell,blocking: (4, 100000, 92, 84, 72.52882821734931, 27.17391304347826, 1.4025513570382959e-18)
N,users,sites,m,capacity,load/cell,blocking: (7, 1000, 23, 48, 38.39164117230078, 1.0869565217391304, 1.4866758266130586e-60)
N,users,sites,m,capacity,load/cell,blocking: (7, 1000, 92, 48, 38.39164117230078, 0.2717391304347826, 4.2401489278533135e-89)
N,users,sites,m,capacity,load/cell,blocking: (7, 10000, 23, 48, 38.39164117230078, 10.869565217391305, 8.388480604414069e-17)
N,users,sites,m,capacity,load/cell,blocking: (7, 10000, 92, 48, 38.39164117230078, 2.717391304347826, 3.6749198106731e-42)
N,users,sites,m,capacity,load/cell,blocking: (7, 20000, 23, 48, 38.39164117230078, 21.73913043478261, 4.4929444957310 96e-07)
N,users,sites,m,capacity,load/cell,blocking: (7, 20000, 92, 48, 38.39164117230078, 5.434782608695652, 6.831870256922693e-29)
N,users,sites,m,capacity,load/cell,blocking: (7, 100000, 23, 48, 38.39164117230078, 108.69565217391305, 0.5652886803903001)
N,users,sites,m,capacity,load/cell,blocking: (7, 100000, 92, 48, 38.39164117230078, 27.17391304347826, 8.789429463380738e-05)
N,users,sites,m,capacity,load/cell,blocking: (12, 1000, 23, 28, 20.150377704053582, 1.0869565217391304, 1.1421964145843877e-29)
N,users,sites,m,capacity,load/cell,blocking: (12, 1000, 92, 28, 20.150377704053582, 0.2717391304347826, 3.581833961698923e-46)
N,users,sites,m,capacity,load/cell,blocking: (12, 10000, 23, 28, 20.150377704053582, 10.869565217391305, 6.444800035458806e-06)
N,users,sites,m,capacity,load/cell,blocking: (12, 10000, 92, 28, 20.150377704053582, 2.7173913043478826, 3.104360910043118 4e-19)
N,users,sites,m,capacity,load/cell,blocking: (12, 20000, 23, 28, 20.150377704053582, 21.73913043478261, 0.035711341422144 61)
N,users,sites,m,capacity,load/cell,blocking: (12, 20000, 92, 28, 20.150377704053582, 5.434782608695652, 5.503817643776413e-12)
N,users,sites,m,capacity,load/cell,blocking: (12, 100000, 23, 28, 20.150377704053582, 108.69565217391305, 0.7454914493631855)
N,users,sites,m,capacity,load/cell,blocking: (12, 100000, 92, 28, 20.150377704053582, 27.17391304347826, 0.1213325942294988 4)
Sanity N=7,n=4: 18.66287339084195 dB
>>>
```

TASK 9 – Record and Present Experimental Performance 3 Marks

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

Answer:

For each strategy and test configuration, record: mean SIR, 5th-percentile SIR, probability SIR falls below 12 dB and 18 dB, Erlang-B blocking probability, channels per cell, number of sites, and simulation run time. Use the same seed and sample count to make comparisons fair. The following tables provide the analytical benchmark and the expected report format; the Python code above generates the position-dependent Monte-Carlo values.

Strategy/configuration	**Channels/cell**	**SIR reference**	**2% GoS traffic/cell**	**Sites for 100k example**
Fixed reuse N=7, omni	48	12.4 dB (n=2.5); 18.7 dB (n=4)	38.39 E	92 after one split
Sectorised N=7, 120°	48 per cell, sectorised	Idealised +4.8 dB vs six-interferer model	38.39 E (channel count unchanged)	Potentially fewer interference-driven sites; traffic still governs
Sectorised N=7, 60°	48 per cell, sectorised	Idealised +7.8 dB vs six-interferer model	38.39 E (channel count unchanged)	Hotspot sectors where antenna design permits

Hierarchical N=7 + DCA	48 nominal/cell	Position/antenna dependent	38.39 E per child cell	Selective; 92 if uniformly split

Recommended plots: (1) SIR cumulative distribution functions for each strategy; (2) blocking probability against offered load per cell; (3) Erlang capacity per cell against N; (4) required site count against subscriber population; and (5) event-zone capacity before and after temporary cells. A plot should label n, N, sector count, sample size and random seed. Do not report invented simulation outputs: execute the supplied program and paste its actual output into the final experimental-results table.

TASK 10 – Compare Theory with Simulation **4 Marks**

Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

Answer:

The analytical first-tier model assumes six equal-distance interferers and a mobile at a representative cell-edge distance. Monte-Carlo simulation samples different mobile positions, so its SIR distribution has a mean and a lower 5th percentile rather than one value. The 5th percentile is the more useful coverage-risk indicator because it exposes

cell-edge and corner users. Sectorised simulation should improve SIR if the assumed directional filtering is valid; actual antenna side lobes will make the gain smaller than the idealised 4.8 dB or 7.8 dB.

Erlang-B theory and the program should agree closely when both use the same channel count and offered traffic. Differences in a live network arise because arrivals may not be Poisson, call durations may be correlated, users may retry blocked calls, data traffic is bursty, and resource scheduling differs from circuit-switched voice. Integer rounding of 336/N also means some channels may remain unused in a cluster.

Other SIR discrepancies arise from shadowing, building penetration, finite rather than infinite co-channel tiers, antenna patterns, power control, unequal base-station distances, terrain and the random distribution of users. The equidistant-interferer approximation may overstate or understate SIR depending on location. A full simulator should include multiple tiers and measured radio maps.

Decision: fixed N=7 is a strong baseline because it meets the 12 dB target in the open-campus ideal model and approaches 18 dB in dense n=4 conditions. Sectorisation improves the interference margin in corridors and hotspots. The hierarchical macro–micro strategy scales best overall because it can add capacity where demand occurs without permanently shrinking every cell. Its advantage is economic and spatial, not an increase in the licensed spectrum.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

Answer:

Recommended deployment: a hierarchical macro–micro network using N=7 as the baseline reuse plan, 48 duplex traffic-channel equivalents per cell, 0.45 km macrocell radius initially, and 0.225 km child-cell radius after one split in high-demand areas. Use 120° sectorisation at major macro sites and 60° sectorisation only where measured interference and directional demand justify the additional antenna and optimisation cost. Adopt 12 dB as the minimum digital-style SIR planning target; use 18 dB as a stretch target for dense high-interference zones.

At 1,000–20,000 subscribers, 23 macrocells provide 883 Erlangs of aggregate 2%-blocking capacity, subject to local hotspot checks. At 100,000 subscribers, one four-way split gives 92 cells and approximately 3,532 Erlangs of aggregate capacity against 2,500 Erlangs offered. Each child cell carries about 27.17 Erlangs on average, below the 38.39 Erlang design capacity. The nominal capacity margin is approximately 41% above offered traffic.

This design is preferable to pure fixed reuse because it responds to non-uniform traffic; preferable to sectorisation alone because sectorisation does not add traffic channels; and preferable to blanket microcell deployment because it limits extra sites to overloaded zones. DCA can borrow idle channels under interference constraints, improving trunking efficiency, but must not violate the reuse plan. The design remains

practical beyond 100,000 users only with ongoing traffic measurement and further selective splitting or additional small cells. It is not an unlimited-capacity solution.

Limitations: the hexagonal grid is an abstraction; site locations, building penetration, antenna patterns, backhaul and power constraints require field validation. Erlang-B is a voice loss model and is not a complete model for packet data. The final channel plan must be validated using a detailed RF planning tool and measured drive-test data.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design **8 Marks**

Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

Answer:

The event adds 25,000 attendees × 0.05 Erlang/person = 1,250 Erlangs in a 1 km² stadium zone for four hours. The event traffic density is therefore 1,250 Erlangs/km². For comparison, the township-wide 100,000-subscriber forecast is 2,500 Erlangs/12 km² = 208.3 Erlangs/km². The stadium event density is about six times the township average. At 20,000 ordinary subscribers, the township average is 500/12=41.7 Erlangs/km², so the event density is 30 times that level.

The assumptions of spatially uniform traffic, steady demand and permanent capacity no longer hold. The demand is concentrated in one square kilometre, predictable, and lasts four hours. Events occur about twice per month, so permanently installing enough capacity for the event would leave expensive resources underused for most of the month. Predictability favours a temporary planned solution, provided setup, testing and backhaul are arranged in advance.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

Answer:

At the 100,000-subscriber design, 92 N=7 cells provide about 38.39 Erlangs each. At the township-average density, 1 km² contains approximately 208.3 Erlangs, equivalent to about 5.4 ordinary cells' engineered traffic capacity. The event adds 1,250 Erlangs in that same square kilometre; the combined local demand is approximately 1,458 Erlangs if the ordinary background demand is included. That is about 38 cell-equivalents at 38.39 Erlangs each.

If the original 92-cell network is spread uniformly, a 1 km² zone contains roughly 92/12=7.67 cells and about 294 Erlangs of engineered capacity. The event alone is 1,250 Erlangs, over four times that capacity; with background traffic it is nearly five times. The first failure is traffic-channel exhaustion and blocking in the stadium cells. Interference may also rise if uncontrolled channel borrowing or ad hoc high-power temporary transmitters are used. The original design cannot be reused unchanged.

C. Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

Answer:

Decision: MODIFY temporarily, retaining the permanent N=7 macro–micro design and adding a pre-planned event overlay. At 38.39 Erlangs per N=7 cell, the event-only minimum is ceil(1,250/38.39)=33 cells. For a robust event plan with background traffic, control overhead, uneven crowds and reserve capacity, deploy approximately 38–40

temporary microcells or cell-on-wheels (COWs) in the stadium zone. A 1 km² area divided into 33 ideal hexagonal cells gives R≈sqrt[1/(33×2.598)]=0.108 km. Forty cells gives R≈0.098 km.

Each temporary cell uses the same N=7 reuse coordination and nominal 48-channel allocation, with carefully planned power reduction to avoid overshooting the small cell. Use 120° or 60° directional sectors where crowd geometry supports it, coordinated channel/resource-block assignment, load balancing and admission control. Maintain macro coverage for mobility and emergency fallback. Handoffs will rise because event microcells are small; mitigate with mobility parameter tuning, overlapping coverage and a temporary event-specific neighbour list.

At 33 cells, nominal capacity is 33×38.39=1,266.9 Erlangs, only slightly above the 1,250 Erlang event load and with little safety margin. At 40 cells, nominal capacity is 1,535.7 Erlangs, providing about 22.9% margin over the event load. If background demand is included, 40 cells are a more defensible starting point; final count should be checked with measured stadium occupancy maps and simulation. Sectorisation primarily improves SIR; the extra microcells provide the traffic capacity.

D. Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

Answer:

A permanent redesign to 40 stadium cells would impose site, radio, power and backhaul costs throughout the month even though the event occurs only about twice monthly. A temporary overlay uses deployable COWs or rapidly installed small cells, with event-specific channel planning and pre-event RF testing. This aligns cost with the four-hour demand window. The analytical evidence is clear: the permanent 92-cell township plan has only about 294 Erlangs of area-proportional capacity in a 1 km² stadium zone, while the event creates 1,250 Erlangs. A 33-cell event overlay is the mathematical minimum under the ideal Erlang-B model; 38–40 cells add practical margin.

The final choice is therefore MODIFY, not RETAIN or full permanent REDESIGN. Retaining the design causes severe blocking; permanently redesigning the whole township is uneconomic for a short, predictable surge. A temporary 38–40-cell event overlay, N=7 reuse coordination, low transmit power, directional sectors and admission control provide the best balance of capacity, SIR, mobility and cost. The simulator should validate the final number using stadium-specific user locations, shadowing, channel occupancy and antenna patterns before each event.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Design Alternative Frequency-Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6

Task 4 – Analyse Best, Average and Worst-Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3
Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4
Task 8 – Implement and Test the Planning Strategies	4
Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	8
TOTAL	**50**